

Relational Affect in Dyadic Interactions

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ABSTRACT

Relational affect is the affective response (encompassing emotion, expression, feeling) that emerges from an interaction between two people. The case study presented here introduces the concept of relational affect through a human perceptual rating task. Forty-five raters watched short video clips of two people interacting and described their perceived emotion of the individuals and that of the overall interaction. Our qualitative analysis of the rater responses showed that raters used a variety of schemes to reason about emotion, including expressions, context, and perceived appraisal of the event. These reasoning schemes were notably different for perceived individual emotion and relational affect. Our findings show that the vocabulary use for relational affect is distinct from that of individual emotion and relational affect as a phenomenon deepens our understanding of social interactions and moves the field a step closer to realizing the goal of fluid interactions between people and technology.

KEYWORDS

Dyadic Interaction, Relational Affect, Affective Computing

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1 INTRODUCTION

Understanding emotions evoked in human-computer interactions (HCI) is key to enabling rich and seamless user experiences [34]. This has never been more critical given the proliferation of social technologies like artificially intelligent chatbots and AI-powered assistants in everyday HCI systems. Consider, for instance, the use of social robots in couples counseling [49], AI chatbots participating in online community spaces [42], software designed to evaluate the effectiveness of teacher student interactions [4], or using AI to track and examine the quality of interaction between customer and service representative [56]. Recognizing and acting upon affective responses plays an increasing role in technologies today.

In the field of affective sciences and affective computing, research on affect tends to largely focus on individual responses including examining a person's facial expressions, speech patterns, or language use. However, in many cases, people experience or express emotions in social interactions, physical or digital, whether between strangers, friends, family, or co-workers. In these social interactions, we propose that affective responses can emerge that are more appropriate to ascribe to the interaction than any one individual participant in that interaction. In this case study, we refer to this phenomenon as “relational affect,” or affective responses (encompassing emotions, expressions, feelings) that emerge from an interaction between two or more people that are more than the sum of its parts (i.e., individuals).

We propose that relational affect is distinct from other previously examined affective processes. First and foremost, relational affect differs from individual affect, particularly in terms of its nature (i.e., a semantic label or description) and time dynamics. Additionally, relational affect can also differ in function compared to individual affect. Self-evaluation strategies or emotion regulation is different in perception of an individual's emotion versus the that of an overall interaction. Relational affect is also distinct from collective emotions. Collective emotions are defined as affective processes that emerge from interactions of multiple people cannot be captured by examining only the responses of individuals [13]. Collective emotions are shared emotional experiences or responses of a group of people [39], whereas relational affect refers to the specific affective responses that emerge from an interaction of two or more people. Additionally, while collective emotions are typically responses to a specific event or situation, relational affect is characterized by the overall dynamics of an interaction. It is also worth distinguishing relational affect from the concept of group-based emotions, which are emotions felt by people based on belonging to a particular social group [45].

Conceptualizing and modeling relational affect can be beneficial in many cases that involve technology being used in social interactions. These cases may arise in interactions between people who are co-located (e.g., measuring engagement between therapist and a client [29], therapeutic alliance), interactions that are computer-mediated (e.g., assessing the quality of a customer care support [21]), or during interactions between humans and digital agents (e.g., experiences with chatbots [55]). In these cases, modeling or probing individual affective experiences may not represent the full affective experience because an affective experience may emerge that results from having interaction itself. For instance, consider a hypothetical situation where two neighbors incidentally run into each other at a social event, exchange smiles and engage in a brief conversation about their day. An observer might describe the interaction as “amicable”, a characterization which may be less

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appropriate to attribute to any one participant in the interaction since both contributed to the emergence of that affective experience. Relational affect becomes even more valuable in cases that involve human-computer interactions where a self-reported affective experience is possible only on the human user end; modeling relational affect can provide insights into the affective performance of digital agents.

Although the term “relational affect” has been invoked in social psychology and philosophy literature in accounts of relational dynamics and interpersonal behavior (see [44, 52] for a few conceptualizations), it has not yet found its way into empirical HCI studies. Empirical studies of relational affect can help develop computational frameworks (e.g., machine perception models) for affective computing in social interaction settings and inform data collection processes including the development of rater guidelines. Furthermore, they can provide useful insights to help create a vocabulary or taxonomy of relational affect concepts and examine the aspects of an interaction (audio, visual or language content) that humans pay attention to when perceiving relational affect.

We present a case study that develops a human-rater study paradigm to elicit relational affect concepts from humans watching a short co-located dyadic interaction. The goal of the human-rater study is to start to understand (1) whether the concepts used to describe perceived relational and individual affective experiences distinct from one another? (2) how perceived relational affective experiences are described alongside individual affective experiences?, and (3) how people reason about relational affect, i.e., what verbal, visual or non-verbal cues are used to describe their perception?

The contributions of this case study are as follows:

- (1) We introduce the concept of relational affect and theorize its role as an emergent property of dyadic interactions
- (2) We design and implement a method to examine perceptions of relational affect in short dyadic interactions
- (3) We identify the unique vocabulary used to describe relational affect phenomenon by analyze the semantic descriptions of relational affect provided by human raters

As HCI technologies approach a more comprehensive understanding of human communication, it becomes increasingly important to promote more fluid interactions between people and technologies. Our work contributes to this goal by improving our understanding of social interactions. Our findings suggest that there are latent properties of dyadic interactions that exist beyond the social cues that individual participants signal. Understanding the nature of these latent properties will be essential to advancing the capabilities of these technologies to perform seamlessly in social interactions. Relational affect is one such property. Understanding how humans perceive and express relational affect can help develop better computer systems that can interact with humans in a more natural and engaging way. This research also has implications for the design of social robots, which are increasingly being used in a variety of settings, such as providing companionship for the elderly [46]. Understanding how humans perceive and express relational affect can enable the design of social robots that are more likely to be accepted and useful to humans.

2 RELATED WORK

In this section, we first discuss a few key theories in the historical development of relational affect concepts from a social psychology perspective. Then, on the use of natural language descriptions to examine the taxonomic space of affective signals followed by the relevant human computer interaction context.

Conceptualization of relational affect: Affect control theory (ACT) [20], first proposed in 1977 [19] is one of the first published works about the collaborative nature of social roles and identities and its intersection with affective concepts such as sentiment. ACT aims to explain human behavior in the context of social interactions. It combined insights from the then predominant affect models: psycholinguistics [33], empirical studies of impression formation [14], and Perceptual Control Theory [37] to form a unified framework. The space of dimensional ratings in ACT is popularly referred to as the “EPA space” and is used to estimate the transient impression evoked by an event along three dimensions: evaluation (good vs. bad), potency (strong vs. weak) and activity (inactive vs. lively / active-passive). An example application of this framework is to predict sentiment in news headlines [2]. Another concept that can be useful to understand relational affect is “interpersonal behavior” i.e., studying the behavior of people in the presence of others. The paper “Interpersonal Circle: A taxonomy for complementarity in human transactions” by Kiesler [24] presented a comprehensive taxonomy of interpersonal behavior. The Interpersonal Circle is a two-dimensional model that organizes interpersonal behaviors into 16 segments, each of which represents a different combination of affiliation and dominance. Affiliation refers to how friendly or hostile someone is, while dominance refers to how controlling or submissive someone is. While the Interpersonal Circle provides valuable insights for relational affect, it has been argued that its reliance on self-report measures, and its static and deterministic view of interpersonal behavior may not capture short-term complex real-world interactions [35]. Relational Models Theory (RMT) [17] is another framing for relational affect in contrast to descriptive concepts of social interactions and relationships such as in ACT. For example, warmth/hostility, strong/weak are often treated as independent and bipolar (expressed the same way by all parties in an interaction) dimensions. In contrast, RMT proposed that these concepts are unipolar in nature for the individuals involved in an interaction. In the present work, we draw from the aforementioned theories of relational affect to study how humans perceive relational affect in short interactions. Unlike these methodologies, we do not fix a taxonomy or a set of dimensions for the raters to describe relational experiences, instead we rely on free-form descriptions from raters.

Natural language descriptions for relational affect: Humans describe social interactions using their own internal representations of what the interactions are about and how they take place. We can then use methodologies such as grounded theory [31] to identify underlying concepts of both the structure (the what) and the function (the how/why) of relational experiences. There are several studies that have applied grounded theory to descriptions of variable framings of relational affect referred to as social episodes, interpersonal situations or situational characteristics. We highlight a few studies relevant to the current work. Forgas et al. [10] is perhaps the earliest

work in this space, which studied the perception of “social episodes”. Here, an episode was defined as a unit of social interaction that has a recognizable beginning and an end. Examples include “playing a game with a child” or “giving a presentation at a meeting”. This study specifically looked at two social groups of about 150 participants including housewives and university students. Representative episodes were presented to the participants and free-form answers were collected using questionnaires. A similar effort was successfully applied to “map out” the space of interpersonal situations [22]. The identified situation patterns included a taxonomy to classify popular interpersonal scenarios such as the prisoner’s dilemma, negotiations and interactions with strangers. Rauthmann et al. [40] identified the dimensions of situational characteristics using a set of nearly 1600 participants. The Riverside Situational Q-sort (a list of 89 items used to describe takeaways from situations used in psychology studies) was administered to collect descriptions of situations. Factor analysis was then applied to identify the key dimensions. Similarly in affective sciences, Cowen and Keltner [8] developed a semantic space approach which aimed to identify taxonomies of emotional expressions based on how people describe emotion-related responses. A curated set of 1500 facial and bodily photographs were presented to over 1700 participants to obtain judgments about perceived emotional expressions from individuals. The free-form responses and appraisal judgments from the participants were then analyzed using grounded theory to develop a taxonomy of perceived emotions. Motivated by these works, our study design involves asking human raters (affect perception) to respond using free-form descriptions. Unlike previous studies which looked at an individual’s response to social scenarios, we focus on understanding human perception of the affective responses attributed to the *dyads* in a social interaction.

Relational affect in human computer interaction: At the time of this case study, a literature search using the term “relational affect” returned no results from the Association for Computing Machinery (ACM) digital library. We therefore broadened our search to include past work examining the affective experiences involving two or more people which includes areas of research in collective emotion and emotion contagion. Regarding collective emotions, researchers have primarily focused on the measurement of collective emotions in public [53] and virtual [11, 12, 18, 30] spaces using novel technological methods [1, 15, 32] and simulation of collective emotions [39]. On emotion contagion, past work has examined how emotions spread across social networking sites [26] and how to simulate the contagious spread of emotions [28, 50, 51, 54]. Our case study builds on this existing body of work by introducing the concept of relational affect and providing an approach to measuring relational affect in the context of dyadic interactions.

3 METHODS

Our definition of relational affect is the set of affective responses that emerge from an interaction between two or more people that are more than the sum of its parts. Based on this conceptualization, a rating task was setup to elicit the following: (1) open-ended description of the perceived emotion i.e., “feeling”. This allows us to explore the relational affect space free of a preconceived taxonomy, (2) one-word descriptions of the feeling of individuals and that of

the overall interaction, allowing us to compare them across the two cases and (3) an open-ended description of the reason for the rater’s response. In this study, we begin by focus on dyadic interactions i.e., interactions between two individuals

3.1 Study design

We use a descriptive study design to examine relational affect of two people interacting with each other. To this end, we setup a rating task to elicit open-ended descriptions of the affective concepts perceived by the raters, both at an individual level as well as that of the “overall interaction” which serves as a proxy for relational affect.

A preview of the rating task as seen by the raters is shown in Fig 1. The task included four main components:

- (1) **Stimulus:** This is a video clip of two people interacting with each other that human raters are asked to view and form an impression. All videos are 10s long and contain two people filmed on a static camera, typically sitting as shown in Fig 1A. The individuals do not change their location for the duration of clip. We excluded any videos where the interaction was extremely staged or the interactants broke the “fourth wall”, i.e., talking to the camera or an individual not seen in the video instead of to each other. The samples included both physically co-located as well as interactions between remote participants. Although the video stimulus is short, it is valid because people can judge others’ affect through a thin-slice of interaction [3]. In addition, we wanted the raters to focus on specific aspects on an interaction. Thus, it helps reduce the cognitive load on raters to watch small segments as opposed to watching and understanding an entire social interaction which can be cognitively demanding.
- (2) **Situation:** As discussed in related work, relational affective experiences are grounded in the situations that the individuals are involved in. For the raters to ground their impression about the overall interaction, we ask the raters to describe the situation with this instruction: “In a few words, describe what happened in the video”.
- (3) **Individual emotion and reasoning:** Given that we constrained our stimulus videos to two people filmed with a static camera and fixed location, we refer to the two individuals as “person on the left” and “person on the right”. Motivated by some of the studies discussed in the related work, we ask the raters to provide free-form descriptions for (a) what they think the person is feeling and (b) how they came to that conclusion. For ease of analysis, we asked the raters to constrain the free-form answers to one word responses. The rating task for reasoning was posed as “Explain why you chose the word above:” as shown in the Figure 1B. We asked the raters to also explain their reasons to understand what aspects of the interaction were used to infer one’s emotion.
- (4) **Relational affect and reasoning:** In order to elicit the relational affective experience perceived by the raters, we posed the question as: “Use one word to describe what you think the overall feeling of the INTERACTION between the two people was:” (See Fig 1C. This allowed us to compare their

(A) Page 1/3: Stimulus



(B) Page 2/3: Describing situation and individual emotion

In a few words, describe what happened in the video:

Use one word to describe what you think the **person on the LEFT** was feeling:
Be specific; Avoid generic words like 'good' or 'bad'

Explain why you chose the word above:

Use one word to describe what you think the **person on the RIGHT** was feeling:
Be specific; Avoid generic words like 'good' or 'bad'

Explain why you chose the word above:

(C) Page 3/3: Describing relational affect

Use one word to describe what you think the overall feeling of the **INTERACTION** between the two people was:
Be specific; Avoid generic words like 'good' or 'bad'

Explain why you chose the word above:

Figure 1: Preview of the rating task as seen by the raters. Includes four components: (1) the stimulus or the video with two persons interacting that the raters view, (2) describe the situation in the video (3) describe what the individual emotion, indexed here as “person on the left” and “person on the right”, and (4) feeling of the overall interaction, which captures the relational affect.

responses with the one-word descriptions of the emotions perceived from the individuals. Similar to the individual emotion questions, we also asked the raters explain their reasoning. Based on the observations from a few pilot runs, we also instructed the raters to avoid quality assessments—such as “good” or “bad”—of the interaction in the one-word descriptions of the individual/relational affect.

3.2 Sample description

We identified a set of 140 videos matching the criteria described in the Stimulus section above. We crowdsourced the rating task described above using Amazon Mechanical Turk (MTurk) with five replications per video. Amazon Mechanical Turk is a popular crowdsource platform that is commonly used by researchers for scaling human perception tasks. The raters were required to be a native U.S. English speakers.

The raters were first pre-qualified through a small pilot rating task to ensure that they understood the task and the responses were specific and relevant to the video being observed. Thus we ended up with 45 unique raters. Each rater was additionally constrained to be able to rate a single video only once and a total of 30 videos out of the 140 video pool to ensure sufficient diversity of responses.

After the task completion and quality control (excluding noisy raters), we obtained a total of 658 ratings. While all 140 videos had at least one rating, 99 videos had all five ratings. On average, the raters took about 5.4 minutes to complete the task. Simple preprocessing was performed to refine the final ratings which included fixing spelling errors and merging different word forms (e.g., “proud” and “pride” were mapped to “proud”). To be faithful the rater responses, we did not do synonym expansion or merging. For example, the labels “unhappy” and “sad” were retained as distinct labels. We

also verified that the individual raters used a sufficiently large vocabulary or set of words to answer the one-word questions. To do this, we computed the ratio of the unique words raters used to the total number of questions answered. The average ratio metric across 45 raters was 7.7 suggesting that most raters provided a sufficiently diverse set of responses for the different stimuli they viewed. Although a single stimulus video was viewed by multiple raters, we chose to analyze the rater responses independent of the stimulus given the subjective nature of the task.

4 ANALYSIS AND RESULTS

We first present an overview of the labels and descriptions provided by the raters. We then present a frequency analysis comparing the concepts used to describe individual emotion and relational affect both at the stimulus level and overall sample level. The frequency analysis enabled us to test the core hypothesis of our study design: the affective concepts used to describe individuals are different from that of the overall interaction.

- (1) **Situation:** Although the stimuli were only 10 seconds in duration, the raters were able to describe the setting of the dyadic interaction sufficiently to ground the subsequent rating tasks related to affect. A few example descriptions of the situation included, “A man gives his opinion about wine while a woman listens”, “A man in a Hawaiian shirt is being interviewed”, “Two attorneys discuss family law” and “discussing sports”.
- (2) **Perceived emotion from individuals:** These one-word descriptions capture the perceived emotions of the individuals in the stimulus videos. Overall, raters used 364 unique words (out of 658 responses) to describe the individual emotions. Examining the individuals separately (referred to as “person



(a) Individual emotion descriptions



(b) Relational affect descriptions

Figure 2: Words used by raters to describe the emotion perceived from the individuals involved in the dyadic interaction and the overall interaction i.e., relational affect. The size of the word is indicative of the frequency of its usage by raters.

Perceived emotion	Reason provided by the rater
Engaged	"She is actively listening and moving the talk forward"
Determined	"She knows what she wants"
Guarded	"He sounds a bit elusive and diplomatic"
Passionate	"She was trying to get his point across intently and with emotion"
Thoughtful	"He voiced his internal deep thought process as he contemplated picking the right jelly bean to eat"
Disinterested	"His smile and head nod looked fake"
Enthusiastic	"She is smiling and talking very passionately about the subject"

Table 1: Examples of reasons provided by raters describing perceived emotion of individuals.

Perceived emotion	Reason provided by the rater
Informative	"They are discussing about a good doctor visit"
Comforting	"The woman talking seems to care about the woman on the right who seems to be sad."
Cordial	"The two men have a very friendly vibe."
Disconnected	"The focus and intensity of the two parties is very dissimilar."
Gratitude	"They are talking about freedom and technology helping them out and are grateful for those things."
Happy	"They are both smiling and appear to be happy and eager to talk."
Tense	"Neither one is accepting of what the other person is saying or trying to say."

Table 2: Examples of reasons provided by raters describing relational affect or the perceived emotion from the overall interaction.

on the left” and “person on the right” in the task), raters used 225 and 242 unique words to describe the person on the left and right respectively. A wordcloud showing the labels provided by raters is shown in Fig 2a. Despite raters being pre-qualified, we noticed that some of the responses for individual emotion included non-affective concepts (e.g., “knowledgeable”), or actions that the person may be doing (e.g., “listening”, “questioning”). We did not remove these terms during preprocessing to be faithful to the rater perception judgment.

- (3) **Reasoning about perceived individual emotion:** A few examples of the perceived emotions from individuals and the corresponding reasons that the raters provided is presented

in Table 1. Qualitatively, in general, raters were more likely to rely on visual, verbal or non-verbal cues to described their reason. For example cues such as “smiling”, “nodding head” or “sounds like”. We also often observed the action being performed by the individuals discussed in the reasons provided.

- (4) **Relational affect descriptions:** These one-word descriptions capture the perceived emotion from the overall interaction which we use as proxy to examine relational affect. Overall, raters used 261 unique words (out of 658 responses) to describe the relational affect. A wordcloud showing the labels provided by raters is shown in Fig 2b. While most descriptions captured the relational nature of the affect, there

were few responses (9 out of 658) that described the setting instead as “interview”. We also noticed that the words used to describe relational affect are distinct from those used to describe individual’s emotions. For example, most frequently used concepts like “cordial” or “friendly” are affective experiences that can be observed only in a social setting.

- (5) **Reasoning about relational affect:** A few examples of the relational affect descriptions and the corresponding reasons that the raters provided is presented in Table 2. Qualitatively, in general, raters mostly relied on *both individuals* in the interaction rather than focusing on a single individual. We found that the reasons were more often closely associated to the topic and the tone of the topic being discussed. Raters rarely reasoned with the expressions of the individuals involved to describe the relational aspects (unlike reasoning about individual’s emotions). Raters also relied on three overarching themes to describe relational experiences: (1) synchronization between the individuals (e.g., “the focus and intensity of the two parties is very dissimilar” to infer “Disconnected”), (2) interaction dynamics (e.g., “neither one is accepting of what the other person is saying or trying to say” to infer “tense”), and (3) appraisal of the situations which involves assessment of the situation or the goals of the topics being discussed (e.g., “they are discussing about a good doctor visit” to infer “Informative”).

One of the main goals of this case study was to examine if the concepts used to describe relational affect are different from those used to describe perceived emotions from individuals. To examine this we conduct a frequency analysis of the corresponding terms raters use to describe an individual’s emotion and the relational affect. First, for each rater response, we computed the frequency at which a rater uses the same word to describe the perceived emotion of either individuals *and* relational affect. In only 26 of the 658 (about 4%) cases was there an overlap in the one-word descriptions of individual and relational affect. Second, we computed the frequency at which there was an overlap between the words used to describe individual emotion and relational affect at the stimulus level, taking into account responses from multiple raters for a given sample. In this case 31 out of 140 videos showed an overlap. In other words, in about 78% of the stimuli relational affect descriptions were different from descriptions of the individual emotion. This suggests that the raters used different concepts to describe the two aspects of perceived emotion in a majority of the cases. Finally, we performed the frequency analysis of the co-occurring terms along three axes: (1) words used to describe the person on the left in the interaction video, (2) words used to describe the person on the right in the interaction video, and (3) words used to describe the relational affect. The frequency analysis table is shown in Table 3. In general the terms used to describe relational affect are unique and different than that of individual emotion. Both response-level and sample-level frequency analysis support our hypothesis that relational affect is distinct from that of individual emotion, i.e., the collective emotion (overall interaction) is greater than the sum of its parts (individuals in the interaction).

5 DISCUSSION

A primary use of affective computing technologies has been to understand and evaluate a wide variety of dyadic interactions such as patient-doctor relationships [43] to customer care and call center representatives [27]. Our work shows that there may be an added benefit to analyzing these interactions in whole, over and above, understanding the responses of individuals in these interactions. By analyzing the interaction as a whole, we show evidence for relational affect, or emergent affective responses that are more attributable to interaction than any individual participant in the interaction.

Recognizing the presence of relational affect may therefore help technologies capture a more complete and nuanced understanding of the affective experience present in dyadic interactions. To illustrate, consider the value of evaluating the relational affective experience in an interaction between patient and doctor. Politeness norms [6] may encourage both parties to express neutral to positive emotions individually despite how the overall interaction might feel. What may be more informative then is to evaluate for the potential presence of negatively- (“tension”) or positively-valenced (“rapport”) relational affect. Such an additional evaluation might reveal a more complete view of how well the interaction is proceeding, enabling more timely and relevant responses. Scenarios such as the patient-doctor conversation generalizes to other dyadic interactions where social norms may operate to potentially obfuscate a more nuanced affective dynamic. Models that recognize and understand relational affect have the potential to make human-computer interactions more fluid by improving technology’s ability to recognize and understand of social dynamics

Our case study also presents a first step toward developing models that recognize relational affect through the development of a rater protocol. The protocol entailed asking people to provide free-form description to what they believed each person in the interaction was feeling. Subsequently, they were asked to provide a free-form description of the overall feeling of the interaction between the two people. Results from ratings revealed that these simple instructions were sufficient for enabling raters to provide unique relational affect labels. Perhaps asking raters to describe the emotional expressions of each individual participant allowed them to reflect on the broader affective responses associated with the overall interaction. Future work could build on this protocol by administering it to a wider array of dyadic interaction stimuli and a wider rater pool demographic to enhance representativeness

Affective experiences are subjective in nature. The human rating task presented in this case study was aimed at prototyping relational affect in dyadic interaction settings, consistent with past works that leveraged human raters for labeling emotion-related concepts [8, 47]. In our early pilot studies, when the raters were given minimal instructions for describing individual emotion/ relational affect, they often responded with labels of “good” and “bad”, describing the *quality* of the interaction or individual behavior. This is consistent with one of the measurement axes, called “evaluation” as proposed by Affect Control Theory. When we further instructed the raters to be specific about the one-word descriptions, we observed more nuance and diversity in how raters described the emotional concepts.

	Person on the Left	Person on the Right	Relational affect
Person on the left	98	61	24
Person on the right	61	113	26
Relational affect	24	26	169

Table 3: Frequency analysis of the terms co-occurring between perceived individual emotion (person on the left/right) and relational affect. The diagonal frequencies show the unique number of terms used to describe the corresponding row/column. While the number shared concepts between individual emotions is higher (61), only at most 26 emotion terms are shared with perceived individual emotion. In addition, there is a larger number of unique terms used to describe relational affect.

In human rating tasks related to affective computing signals for individuals, it is a popular practice to use pre-defined taxonomies to confine the label space, typically for developing subsequent machine learning models (for example, see [7, 41]). In contrast, we sought to examine a novel concept of emotion grounded to that of an interaction rather than an individual. Hence, we chose a free-form response style from the raters. As a consequence, some of the responses for relational affect were not strictly emotional concepts. For example, responses such as “casual” and “professional” (see Fig. 2b) suggested a larger set of relational experiences.

Our qualitative analysis of reasons provided by raters for individual emotion was consistent with prevalent theories in affective sciences. For example, raters relying on individuals’ expressions (visual cues such as smiles, verbal tone or non-verbal cues) to perceive emotion is consistent with a significant literature in affective sciences and basic emotion theory [23]. In addition, raters also considered the social context in which the emotions were being expressed. For example, reasons such as “She is actively listening and moving the talk forward” inferring “Engaged” accounted for the situational context. These observations are consistent with theory of constructed emotion [5]. Additionally, the reasons provided by raters for relational affect appear to be consistent with relational phenomena such as interaction dynamics [36], dyadic synchrony [48] and appraisal theory of emotion [16]. Furthermore, our qualitative comparison of the reasons that raters provided for describing individual versus relational affect (see Tables 1 and 2) suggest different human mental models underlying corresponding inferences. For example, raters considered perceived appraisals of the situations more often to reason about relational affect than individual emotion.

In comparing how raters described individual vs. relational affect, a key takeaway from our case study was that *Raters rarely reasoned about relational affect with the expressions of the individuals involved, unlike reasoning about an individual’s emotions*. Raters tended to rely on situational aspects or the interplay between the two individuals in the interaction to infer relational affect. This finding supports and highlights the need that recent studies in affective sciences (e.g., [38]) and affective computing (e.g., [25]) have been focusing on in regard to including contextual factors for modeling emotion perception tasks. Finally, this is a case study examining around 700 rater responses on a set of 140 of video stimuli. The objective here was to develop a working definition of relational affect and prototype it with a human rater task to understand how it compares with emotional concepts grounded to an individual. While the methodologies developed here can be applied to other forms of relational affect phenomena such as collective emotion in larger

groups, or longer conversations or multiparty interactions, further development is necessary.

6 FUTURE DIRECTIONS

For ease of analysis and developing a proof of concept, we restricted the rater responses for individual emotion and relational affect to one-word responses. While this allowed us to easily compare the vocabulary spaces of the two emotion types, it also restricted the amount of details and nuance that raters could describe. One future direction in this regard is to allow raters to use free-form descriptions and then use sentence embeddings (e.g., BERT [9]) to operate in the dimensional space rather than a discrete taxonomy space. The stimulus videos used in this study were not entirely naturalistic conversations or interactions. They often involved users who knew a priori that their interactions were being filmed or had an expectation thereof. Using naturalistic interaction stimuli may offer additional insights into relational affective phenomena. Our preliminary results also suggest a taxonomy of relational affect that is significantly different from that of the individual emotion, possibly scoped broader than strictly affective experiences. Scaling up the proposed study is a promising direction to develop such a taxonomy.

7 CONCLUSION

We present a case study to examine affective phenomenon that is grounded in an overall interaction or conversation, referred to as *relational affect* in comparison with individual affect. Toward this goal, we developed a human rating task using video stimuli of dyadic interactions to obtain free-form description of raters’ perception of the emotions being expressed by the individuals in the interaction and the overall interaction itself, a proxy for relational affect. Frequency analysis of the co-occurring individual and relational affect terms on a sample of 658 responses for 140 stimuli showed that raters used a distinct vocabulary to describe relational affect compared to that of the individual emotions. Qualitative analysis of the reasoning task suggest that human raters used a variety of schemes to reason about individual and relational emotion, including expressions, context, and perceived appraisal of the event.

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